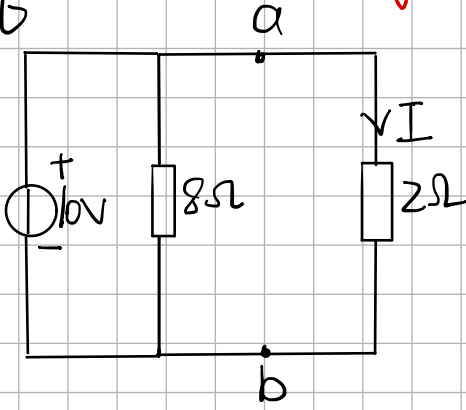


习题 1

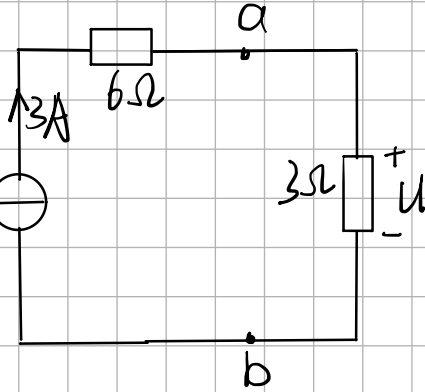
1-6

(a)



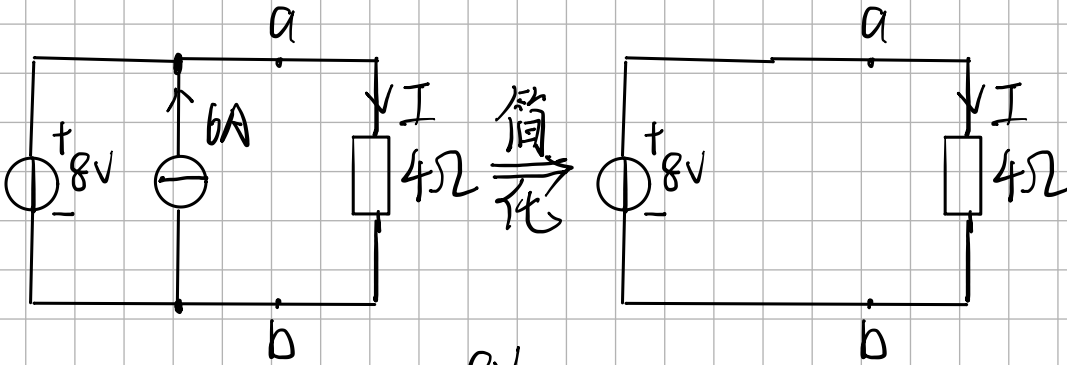
$$I = \frac{U}{R} = \frac{10V}{2\Omega} = 5A$$

(b)



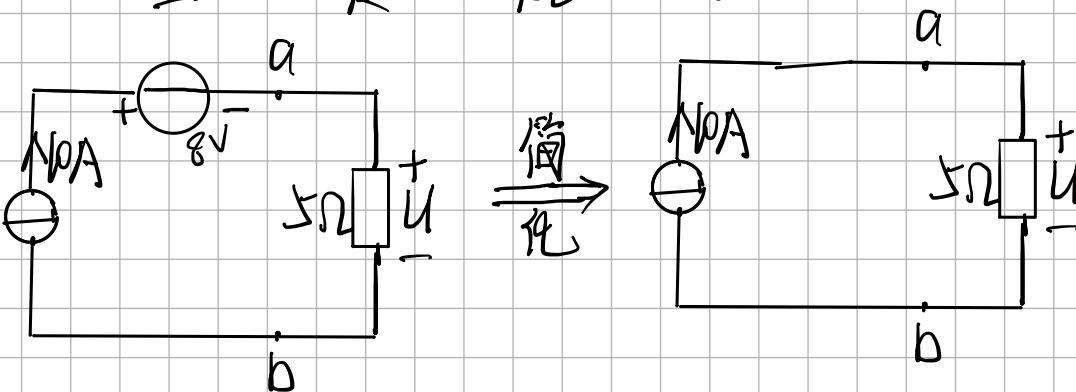
$$U = IR = 3A \times 3\Omega = 9V$$

(c)

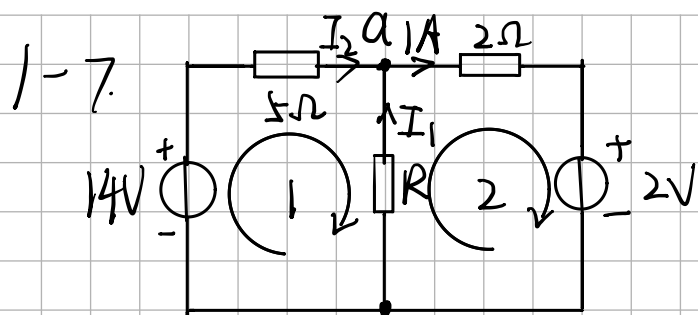


$$I = \frac{U}{R} = \frac{8V}{4\Omega} = 2A$$

(d)



$$U = IR = 10A \times 5\Omega = 50V$$



如图 1、2 两个环路，设节点 a ，流过 R 和 5Ω 电阻的电流为 I_1, I_2

由环路 1、2 列出 KVL 方程

$$5I_2 - I_1 R - 14V = 0 \dots\dots ①$$

$$I_1 R + 2V + 2V = 0 \dots\dots ②$$

由节点 a 列出 KCL 方程

$$I_1 + I_2 - 1A = 0 \dots\dots ③$$

联立 ① ② ③ 方程得

$$\begin{cases} R = 4\Omega \\ I_1 = -1A \\ I_2 = 2A \end{cases}$$

得电阻 R 的阻值为 4Ω

设 $14V, 2V$ 电源, $5\Omega, 2\Omega$ 电阻, R 电阻吸收的电功率分别为 $P_{14V}, P_{2V}, P_{5\Omega}, P_{2\Omega}, P_R$

$$P_{14V} = -14I_2 = -28W$$

$$P_{5\Omega} = 5I_2^2 = 20W$$

$$P_{2\Omega} = 2I_2^2 = 2W$$

$$P_{2V} = 2 \times 1 = 2W$$

$$P_R = RI_1^2 = 4W$$

$$P_{14V} + P_{5\Omega} + P_{2\Omega} + P_{2V} + P_R = 0W$$

证毕

★ 关联参考方向：吸收
★ 非 ... : 发出